

Arthur Le Saux, Ph.D.

CEA, AIM $\diamond$ CNRS $\diamond$ Université Paris-Saclay $\diamond$ Université Paris Cité

alesaux@proton.me $\diamond$ <https://arthurlesaux.github.io/>

Fluid dynamics $\diamond$ Stellar Physics $\diamond$ Atmospheric sciences $\diamond$ Numerical simulations

POSITION

Postdoctoral Research Fellow (1 year renewable ERC contract) *Octobre 2024 - Present*

AIM, CEA, Université Paris-Saclay

Funded by the [4D-STAR](#) ERC Synergy grant (PIs: Profs. Drs. C. Aerts, S. Mathis, M. Rieutord and A. Dotter)

Impact of rotation and magnetism on asteroseismic properties of stars. Development of multidimensional stellar evolution code.

PUBLICATIONS & PRESENTATIONS (LIST P. 3)

12 Papers in peer-reviewed scientific journals: 3 first author, 5 major contributions and 4 co-author contributions (Citations = 113, h-index = 6, i10 index = 4).

Google Scholar profile: https://scholar.google.com/citations?hl=fr&user=w_jPdt4AAAAAJ

10 Conference proceedings

8 Press releases on artistic collaboration

28 Oral Presentations: 13 in international conferences and 7 invited

10 Posters: 8 in international conferences

EDUCATION

Ph.D. in Physics & Astronomy, University of Exeter *June 2023*

Thesis: Understanding mixing processes in stars using hydrodynamic simulations

Master of Science (MSc), Observatoire de Paris *August 2019*

Astronomy, Astrophysics & Space Engineering

Specialisation: Stellar physics, Asteroseismology, Numerical Methods, Data Science

Master of Engineering (MEng), Ecole Centrale de Lyon *August 2018*

Specialisation: Hydrodynamics, Nanotechnologies and Biotechnology

Bachelor of Science, Sorbonne Université *June 2015*

Mechanical Engineering

Specialisation: Fluid dynamics, Acoustics, Thermal Physics

French Baccalaureate, Lycée Alphonse Daudet *June 2011*

Scientific, with Highest Honours

Specialisation: Life and Earth Science and Mathematics

RESEARCH EXPERIENCE

Postdoctoral Research Fellow (2 years contract) *June 2023 - October 2024*

Laboratoire de Météorologie Dynamique, Sorbonne Université

Funded by the [ANR SOUND](#) grant (PI: Dr. S. Guerlet)

Modelling the atmospheric dynamics of Neptune and Uranus using a Global Climate Model

Postgraduate Research Student *Sept. 2019 - June 2023*

Physics & Astronomy, University of Exeter

Co-direction: CRAL, Ecole Normale Supérieure, Lyon, Université de Lyon

Supervisors: Prof. Isabelle Baraffe & Dr. Thomas Guillet

Study of waves and convective penetration in stellar interiors using the fully compressible time implicit hydrodynamical code MUSIC

Master's thesis (5 months)	<i>March 2019 - August 2019</i>
AIM, CEA, Université Paris-Saclay	
Supervisors: Dr. Rafael A. García & Dr. Lisa Bugnet	
Identification and classification of pulsating stars by combining classical asteroseismology and machine learning methods	
Observation Trainee (1 week)	<i>March 2019</i>
Instituto de Radioastronomía Milimétrica, Granada	
Supervisors: Dr. Pierre Guillard	
Turbulence and proto-stellar outflows mapping in L1251-A	
Master's thesis (6 months)	<i>March 2018 - August 2018</i>
Laboratoire de Météorologie Dynamique, Sorbonne Université	
Supervisors: Dr. Sébastien Lebonnois	
Study of large-scale gravity waves in the deep atmosphere of Venus	
Research Intern (3 months)	<i>April 2017 - July 2017</i>
Physics & Astronomy, University of Exeter	
Supervisors: Prof. Isabelle Baraffe	
Theoretical and numerical approach of hydrodynamic processes in stellar interiors	
Research Project	<i>Sept. 2017 - June 2018</i>
Institut des Nanotechnologies de Lyon	
Supervisors: Dr. Bertrand Vilquin & Dr. Pedro Rojo Romeo	
Theoretical and practical study on hyperbolic metamaterials	

GRANTS & AWARDS

Grant from SF2A to support an event to promote astronomical culture (2100\$)	<i>January 2025</i>
CNES Postdoctoral Fellowship (Declined because of position at CEA-Saclay)	<i>Spring 2024</i>
Kavli Summer Program in-person workshop, Göttingen, Germany (1000\$)	<i>Fall 2022</i>
Kavli Summer Program Fellowship (Awarded to 8 fellows/ year) (2000\$)	<i>Spring 2021</i>
1st Prize Postgraduate Conference Presentations, Exeter, UK (60\$)	<i>Winter 2021</i>
Artistic residency, Le Port des Créateurs, Toulon, France (7000\$)	<i>Winter 2019</i>
College funded PhD Studentship (3.5 years), University of Exeter, UK (20000\$ /yr)	<i>Spring 2019</i>
Observation time (36h), IRAM 30m telescope, Granada, Spain	<i>Spring 2019</i>

STUDENT SUPERVISION

Kanah Smith , 4 month (40% supervision), in the context of the Simons-NSBP Scholars Program , hosted by the CCA, Flatiron Institute, New-York, USA, + 3 years (10% supervision) during PhD thesis at ISTA, Vienna, Austria Co-supervisor: Dr. Lisa Bugnet	<i>Since 2022</i>
Armand Leclerc , 6 month (60% supervision), University of Exeter, UK, Co-supervisor: Prof. Isabelle Baraffe	<i>Summer 2021</i>
Benoit Gay , 6 month (40% supervision), University of Exeter, UK, Co-supervisor: Prof. Isabelle Baraffe	<i>Summer 2020</i>

TEACHING

Teaching Assistant at the MESA summer school 2025 , Leuven, Belgium	<i>July 2025</i>
Teaching Assistant at the MESA summer school 2023 , Budapest, Hungary	<i>August 2023</i>
Demonstrator for 2nd year Physics problem class in Quantum Mechanics, Electromagnetism, Thermal Physics and Condensed Matter (96 hr/yr)	<i>2020 - 2023</i>
Private lessons to high school students (2h/ week)	<i>2010-2015</i>

LANGUAGES

Human	French (native), English (B1, fluent), Spanish (conversational), Italian (beginner)
Programming (Good expertise)	Python, Bash, Fortran 77/95/08, HTML, Excel
Programming (Notions)	MATLAB, R
Python Packages	Pandas, Matplotlib, Numpy, Scipy, h5py, xarray

OUTREACH

UPCOMING: <i>Arts and Sciences Fair</i> , Le Port des Créateurs, Toulon, France	<i>Octobre 2025</i>
<i>Pint of Science festival</i> , Les Marquises, Paris, France	<i>May 2025</i>
Invited conference on the relationship between art and science, with L. Fourdrinier at MO.CO, Montpellier Contemporain, France	<i>March 2025</i>
<i>Astronomy on Tap</i> , Elephant & Castle, Lyon, France	<i>June 2024</i>
<i>The music of stars</i> , public talk at The Cabin, Praha, Czech Republic	<i>May 2023</i>
Scientific mediation action for high school students, organised by Le Cercle FESR and Declics, Lycée Du Parc and Lycée Saint-Just (Lyon), and Lycée Henri Bergson (Paris)	<i>2022 - 2023</i>
Invited conference on the relationship between art and science, MUREX festival	<i>July 2021</i>
<i>Street confs</i> , Presentation of scientific research and stellar physics to the public in the streets of Marseille	<i>June 2019</i>

ARTISTIC COLLABORATION

Active collaboration with [Léo Fourdrinier](#), plastic artist working in Toulon (France). Through regular discussion, Léo reinterprets my research on stars and planets for his plastic work.

<i>Poems hides theorems</i> , Leo Fourdrinier solo exhibition, reinterpretation of works from artistic collaboration, Galerie les Filles du Calvaire (Paris)	<i>2024</i>
<i>Art Genève</i> , Contemporary and modern art international fair, one piece from artistic collaboration, Palexpo (Geneva)	<i>2024</i>
<i>La lune dans un oeil et le soleil dans l'autre</i> , Solo art exhibition of Léo Fourdrinier, with several pieces from artistic collaboration, CACN , Nîmes, France.	<i>2022</i>
<i>PULSE</i> , Art Exhibition on the theme of asteroseismology, L'Axolotl Gallery (Toulon)	<i>2020</i>
Collaboration initiated during the Booster #4 artistic residency hosted at Le Port des Créateurs (Toulon)	<i>2019</i>

CONFERENCE ORGANISING

Chair and organizer of the workshop <i>Transport processes in stars and planets: observational constraints and theoretical challenges</i> at the Journées de la SF2A 2025, Toulouse, France	<i>1st-4th July 2025</i>
Organiser for a Parallel Session at the National Astronomy Meeting 2023	<i>3rd-7th July 2023</i>
Organiser for the EMPS PGR conference 2022 (University of Exeter) and designer of the conference website	<i>4th-5th April 2022</i>

PUBLICATIONS LIST

◦ Publications in peer-reviewed scientific journals (12)

First author publications (3)

1. **Le Saux, A.**, Leclerc, A., Laibe, G., et al., *A one-hour topological oscillation explores the Sun's core* (Accepted for publication in ApJ Letters, [arXiv:2506.09572](#))

2. **Le Saux, A.**, Baraffe, I., Guillet, T., et al., *Two dimensional simulations of internal gravity waves in a 5 solar mass star zero-age-main-sequence model* (2023), MNRAS 552, 2
3. **Le Saux, A.**, Guillet, T., Baraffe, I., et al., *Two-dimensional simulations of solar like models with artificially boosted luminosity. II. Impact on internal gravity waves* (2022), A&A 660, A51

Major contributions (5)

1. Buldgen, G., Canocchi, G., **Le Saux, A.** et al., *The Future of Solar modelling: requirements for a new generation of Solar models.*, (Accepted for publication in Solar Physics, [arXiv:2506.14514](https://arxiv.org/abs/2506.14514))
2. Morton, J., Guillet, T., Baraffe, I., Morison, A., **Le Saux, A.** et al., *Limitations of Tracer Particle Methods for Characterising Mixing by Internal Gravity Waves in Massive Main-Sequence Stars*, (Accepted for publication in MNRAS)
3. Morison, A., **Le Saux, A.**, Baraffe, I., et al. *Effect of stratification atop the convective core of 5 solar mass main-sequence star* (2024), MNRAS 532, 2
4. Baraffe, I., Constantino, T., Clark, J., **Le Saux, A.** et al., *Local heating due to convective overshooting and the solar modelling problem* (2022), A&A 659, A53
5. Vlaykov, D. G., Baraffe, I., Constantino, T., Goffrey, T., Guillet, T., **Le Saux, A.** et al., *Impact of radial truncation on global 2D hydrodynamical simulations for a Sun-like model* (2022), MNRAS 514, 1

Co-author contributions (4)

1. Stuck, M., Pratt, J., [...], **Le Saux, A.**, *Convective shells in the interior of Cepheid variable stars: overshooting models based on hydrodynamic simulations*, (Accepted for publication in A&A, [arXiv:2505.04900](https://arxiv.org/abs/2505.04900)),
2. Dethero, M-G., Pratt, J., [...], **Le Saux, A.** et al., *The shape of convection in 2D and 3D global simulations of stellar interiors*, (2024), A&A 692, A46
3. Baraffe, I., Clarke, J., [...], **Le Saux, A.** et al., *A study of convective core overshooting as function of stellar mass based on twodimensional hydrodynamical simulations* (2023), MNRAS, 519, 4
4. Baraffe I., Pratt J., [...], **Le Saux A.** et al., *Two-dimensional simulations of solar like models with artificially boosted luminosity. I. Impact on convective penetration* (2021), A&A 654, A126

◦ **Conference Proceedings (10)**

1. **Le Saux, A.** et al., *Exploring the atmospheric dynamics of the Ice Giants: A General Circulation Model Approach with a Parametrization of Convection* (2024), Europlanet Science Congress 2024
2. Carrión-González, O., [...], **Le Saux, A.** et al., *Measuring Neptune's temperature maps and CO abundance with ALMA* (2024), Europlanet Science Congress 2024
3. Smith, K., M., **Le Saux, A.** & Bugnet, L., *Unveiling stellar nature through oscillation pattern recognition*, 8th TESS/15th Kepler Asteroseismic Science Consortium Workshop
4. **Le Saux, A.** et al., *Investigating Atmospheric Dynamics of Uranus and Neptune: A General Circulation Model Approach with a Parametrization of Convection* (2024), EGU General Assembly 2024
5. N. Clément, [...], **Le Saux, A.** et al., *Convective storms and methane clouds on Uranus & Neptune: sporadicity and latitudinal variations revealed by a 3D cloud-resolving model* (2024), EGU General Assembly 2024
6. **Le Saux, A.**, *Exploring waves properties with multi-dimensional fully compressible hydrodynamical simulations: from solar type stars to intermediate-mass stars* (2023), American Astronomical Society Meeting 241

7. Smith, K., **Le Saux, A.**, Bugnet, L., *Asteroseismology: Unveiling Stellar Nature Through Oscillation Pattern Recognition* (2023), American Astronomical Society Meeting 241
8. **Le Saux, A.** et al., *Modelling waves in solar-like interiors with the MUSIC code* (2022), The 21st Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun
9. **Le Saux, A.**, Bugnet, L., Mathur, S., et al. *Automatic classification of K2 pulsating stars using Machine Learning* (2019), SF2A 2019 Proceeding
10. Breton, S., [...], Le Saux, A. et al. (2019), *Determining surface rotation periods of solar-like stars observed by the Kepler mission using machine learning techniques* SF2A 2019 Proceeding
11. Lebonnois, S., Garate-Lopez, I., **Le Saux, A.** & Plougonven, R., *Wave activity below the clouds of Venus with the IPSL Venus GCM* (2019), 21st EGU General Assembly

◦ **Press releases on artistic collaboration (8)**

1. *La sérendipité post-antique de Léo Fourdrinier*, The Steidz, Oct. 12th, 2024
2. *Léo Fourdrinier at the Centre d'Art Contemporain de Nîmes*, Artmirror, Feb. 9th, 2022
3. *Nîmes : l'antiquité et les étoiles dans les rêveries de Léo Fourdrinier au nouveau Centre d'Art contemporain* (French), Midi Libre, Oct. 19th, 2021
4. *PULSE, con obra de Léo Fourdrinier* (Spanish), [Anti]Materia, April 2nd, 2020
5. *PULSE, Léo Fourdrinier At L'axolotl*, Toulon, France (English), O Fluxo, March 2020
6. *Exposition PULSE de Léo Fourdrinier* (French), Revue Point Contemporain, March 2020
7. *Léo Fourdrinier - La tête dans les étoiles* (French), Cité des Arts, Feb. 3rd, 2020
8. *Voyage dans le vide sidéral au Port des créateurs* (French), Var Matin, Jan. 25th, 2020

ORAL PRESENTATIONS & POSTERS

◦ **Oral presentations (28)**

Invited presentations (4)

1. Turbulence Day at ENS de Lyon, June 2024, *Lyon, France*
2. Beyond Boussinesq workshop, October 2023, *Lyon, France*
3. Future of solar modelling workshop, September 2023, *Sierre, Switzerland*
4. Fluid Dynamics of the Sun and stars workshop, November 2022, *Göttingen, Germany*

Invited seminars (3)

1. Institute of Science & Technology, February 2024, *Klosterneuberg, Austria*
2. LESIA, Observatoire de Paris, Université PSL, May 2023, *Meudon, France*
3. Observatoire de Genève, December 2022, *Versoix, Switzerland*

Seminars (5)

1. Laboratoire Univers et Particules de Montpellier, March 2025, *Montpellier, France*
2. Institut de Recherche en Astrophysique et Planétologie, March 2023, *Toulouse, France*
3. Astrophysics group (Seminar with L. Fourdrinier), February 2023, *University of Exeter, UK*
4. Good Vibrations seminar series, April 2022, *Online*
5. Astrophysics group, December 2021, *University of Exeter, UK*

Contributed talks (16)

1. EPSC Assembly, Septembre 2024, *Berlin, Germany*
2. Meeting SF2A 2024, June 2024, *Marseille, France*
3. EGU General Assembly 2024, April 2024, *Vienna, Austria*
4. Perspectives colloquium, French National Program in Stellar Physics, January 2024, *Marseille, France*
5. Generic PCM days, LMD, October 2023, *Paris, France*
6. MUSIC Team Talk, October 2023, *Online*
7. AAS #241 meeting (remote **Dissertation talk**), January 2023, *Seattle, USA*
8. Fluid Dynamics of the Sun and stars workshop, November 2022, *Göttingen, Germany*
9. MUSIC Team Talk, November 2022, *University of Exeter, UK*
10. Angular momentum in stellar systems workshop, September 2022, *CEA, Paris, France*
11. Cool Stars 21 (**Plenary session**), July 2022, *Toulouse, France*
12. French National Program in Stellar Physics workshop: Codes for stellar physics, June 2022, *Observatoire de Paris, Meudon, France*
13. European Astronomical Society meeting 2022, Session 21, June 2022, *Valencia, Spain*
14. MUSIC Team Talk, November 2021, *Online*
15. Kavli Summer Program in Astrophysics, June 2021, *Online*
16. EMPS PGR Conference 2021 (**Best Presentation Award**), March 2021, *Exeter, UK*

◦ Posters (10)

1. *Exploring the influence of small-scale convection parametrisation on ice giant planets atmospheric dynamics*, Parametrizations for Global Dynamical Models, Mars 2025, Les Houches, France
2. *Topological Lamb-like waves could help to probe the innermost regions of the Sun*, Journées SF2A 2024, Marseille, France
3. *Modelling stellar interiors with the MUSIC code*, Journées SF2A 2023, Strasbourg, France
4. *Unveiling stellar nature through oscillations pattern recognition*, EAS General Assembly, 27 June – 1 July 2022, Valencia, Spain
5. *Modelling waves in stellar interiors with the MUSIC code*, TASC6/KASC13 Workshop, 11-15 July 2022, Leuven, Belgium
6. *Two-dimensional simulations of solar like models with artificially enhanced luminosity*, Kavli Summer Program in Astrophysics 2021, June 2021, Online
7. *Two-dimensional simulations of solar like models with artificially enhanced luminosity*, Cool Stars 20.5, March 2020, Online
8. *Internal gravity waves in two-dimensional compressible simulations of solar like models*, PGR conference 2021, 10-11 March 2021, Exeter, UK
9. *Internal Gravity Waves in Stars: Impact of artificially boosted models*, Atelier ExoSystèmes I : Evolution, 28-30 Janvier 2020, Paris, France
10. *Automatic classification of K2 pulsating stars using Machine Learning*, Journées SF2A 2019, Nice, France